

ANKUSH KUMAR MALLICK

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Professional Summary

AI/ML Engineer with hands-on experience building and evaluating machine learning and deep learning models in Python. Worked on end-to-end projects involving data preprocessing, model training, evaluation, and deployment using PyTorch, TensorFlow, and scikit-learn, with a focus on NLP and practical problem-solving.

Education

University of Engineering & Management, Jaipur
Bachelor of Technology in Computer Science Engineering

Aug 2022 – Present
Rajasthan, India

Technical Skills

Languages: Python, C++, C, SQL

DL Topics: ANN, CNN, RNN, LSTM, Transformers, Batch Normalization, Dropout, Backpropagation

NLP Topics: Word Embeddings (Word2Vec), BERT, TF-IDF, Tokenization, Lemmatization

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Streamlit, Flask

Data Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Model Optimization: Hyperparameter Tuning, Early Stopping, Learning Rate Scheduling, Regularization, Cross-Validation

CS Fundamentals: Data Structures & Algorithms (DSA), OOP, DBMS

Projects

ResuScore: AI-Powered Skill Gap Analyzer | *Sentence-Transformers, Cosine Similarity, Streamlit* GitHub

- Developed a full-stack AI application using **Sentence-Transformers** for semantic resume-job description matching.
- Engineered a deep learning pipeline using the **all-MiniLM-L6-v2** model to convert text into 384-dimensional dense vectors.
- Implemented **Cosine Similarity** algorithms to provide a quantitative "Match Score" based on contextual understanding.
- Visualized skill gaps using **Plotly Radar Charts** and deployed on Streamlit Cloud for real-time interaction.

ReviewLens: Hybrid Fake Hotel Review Detection | *NLP, Random Forest, Flask, NLTK* GitHub

- Built a deception detection system achieving **89.7% accuracy** and **0.94 recall** using the Deceptive Opinion Spam Corpus.
- Engineered a **Hybrid Feature Pipeline** merging **TF-IDF** (3,000 features) with **Stylometric Features** (Lexical Diversity).
- Optimized a **Random Forest** architecture to handle high-dimensional feature sets, outperforming baseline models.
- Deployed a real-time **Flask** dashboard with NLTK-based preprocessing and an AI Confidence Score verdict.

Travel Budget Prediction System | *Python, XGBoost, Scikit-learn, Streamlit*

- Developed an ML regression model to predict travel budgets using **Feature Engineering** on user inputs such as travel style and trip duration.
- Applied numeric feature preprocessing and **Hyperparameter Tuning** using Scikit-learn, achieving a **R² score of 0.97** on test data.
- Built an interactive Streamlit web application for real-time budget forecasting and saved models using joblib.
- Maintained a deployment-ready structure with requirements.txt and ensured model robustness through feature selection.

Certifications

Python for Data Science – NPTEL	2024
Fundamental Algorithms: Design and Analysis – NPTEL	2024
The Joy of Computing using Python – NPTEL	2023
Problem Solving Through Programming in C – NPTEL	2023